

What a Single mAP Hides: A Measurement Audit of Six-Class Dental Condition Detection on Panoramic Radiographs

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DentalScan AI technical report

github.com/.../dentalscan-ai

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Abstract

We revisit a six-class dental condition detector trained on 231 annotated panoramic radiographs, previously summarised by a single figure of $\text{mAP}@0.5=0.92$. Decomposing that number reveals that it is not a measurement. Three of the six classes have fewer than ten instances in the validation split and one has zero instances in the test split; the two classes that report near perfect average precision are supported by three and eighteen boxes respectively, and a 95% interval on the rarest class’s recall spans $[0.44, 1.00]$. We report per-class precision, recall and average precision for three detector families trained under an identical recipe, decompose every error into a named category, and show that essentially all of the residual error is missed detections and background false positives—inter-class confusion accounts for roughly one percent. We then specify and implement the protocol the dataset actually supports: grouped, rarity-aware k -fold cross-validation that raises evaluation support for the rarest class from 3 instances to 72, a multi-seed ablation over the augmentation recipe with paired significance testing, and a saliency pipeline scored by pointing-game energy and deletion AUC rather than shown as illustrations. The headline result of the audit is that the largest detector in the comparison costs $4.6\times$ the training time of the best one and does not beat it.

1 Introduction

Object detectors applied to medical imaging are usually reported with a single aggregate number. On a balanced benchmark such as COCO [4], where each class carries thousands of instances, the mean over classes is a reasonable summary. On a small clinical dataset it is not: the mean is dominated by the frequent classes, and the rare classes—which are typically the clinically interesting ones—contribute values estimated from a handful of boxes.

This report audits a working system. **DentalScan AI** detects six conditions (Caries, Infection, Impacted, BDC/BDR, Fractured, Healthy) in panoramic radiographs and is deployed as a browser tool. Its previously reported figure was $\text{mAP}@0.5=0.923$ for a YOLOv10s [11] backbone trained for 250 epochs. We ask three questions:

1. What does that number consist of, per class?
2. Which errors does the model actually make?

3. What would it take to measure this system honestly?

Our contributions are (i) a measurement-validity audit that quantifies why the current split cannot support per-class claims; (ii) a per-class head-to-head of three detector families under an identical recipe, including training cost; (iii) an error decomposition that identifies the dominant failure mode; and (iv) an open harness—cross-validation, multi-seed ablation, bootstrap intervals, and faithfulness-scored saliency—that makes the corrected protocol one command to run.

2 Related work

Dental radiograph detection. Public panoramic datasets remain small. The Tufts Dental Database [6] provides 1,000 panoramic images with expert annotation; the dataset used here [1] provides 231 annotated radiographs across six conditions. At this scale, the gap between reported and achievable performance is usually a property of the evaluation protocol rather than of the architecture—a point argued in general terms for medical imaging by Varoquaux and Cheplygina [10].

Error decomposition. TIDE [2] showed that a scalar mAP conflates qualitatively different failures and proposed sorting false positives into classification, localization, duplicate and background categories. We adopt that taxonomy, adding missed ground truth as a sixth bucket, because on this dataset it turns out to dominate.

Variance. Picard [8] demonstrated that seed-to-seed variation in vision models is large enough to account for many published improvements. On a 23-image validation split the effect is amplified, which motivates our insistence on repeated runs.

Explanation. Grad-CAM [9] and its gradient-free relative Eigen-CAM [5] produce saliency maps for convolutional networks. Saliency maps are frequently shown without evidence that they describe the model; we score ours with a pointing-game energy statistic and a deletion curve in the style of RISE [7].

3 Data and the measurement problem

The dataset comprises 231 annotated panoramic radiographs. The public release ships a split in which the training portion has been offline-augmented $3\times$; Table 1 gives the resulting counts.

Class	Train	Val	Test	Total
Healthy	3293	67	54	3414
Caries	853	14	26	893
Impacted	729	18	12	759
Fractured	526	9	5	540
Infection	172	4	5	181
BDC/BDR	69	3	0	72
Boxes	5642	115	102	5859
Images	558	23	23	604
Radiographs	186	23	23	231

Table 1: Instance counts per split. The training split contains three offline augmented copies of each of 186 source radiographs. Class imbalance in training is 48:1. BDC/BDR has three instances in validation and none at all in test.

Class	Recall	Support	95% interval
Impacted	1.00	18/18	[0.82, 1.00]
Fractured	0.89	8/9	[0.56, 0.98]
Caries	0.79	11/14	[0.52, 0.92]
Healthy	0.64	43/67	[0.52, 0.75]
BDC/BDR	1.00	3/3	[0.44, 1.00]
Infection	0.50	2/4	[0.15, 0.85]

Table 2: Per-class recall of the YOLOv10s model on the validation split, with Wilson score intervals. Four of six intervals are wider than 0.40—wider than any effect an ablation on this dataset could plausibly produce.

Leakage. We first verified that offline augmentation did not contaminate evaluation. Grouping every file by the source radiograph encoded in its filename, the intersection between every pair of splits is empty (0 shared radiographs for all three pairs). The reported numbers are therefore not inflated by leakage—but they are inflated by something else.

The validation split cannot carry per-class claims. Three of six classes have fewer than ten validation instances. Table 2 makes the consequence concrete: even taking the model’s observed per-class recall at face value, the 95% Wilson interval on BDC/BDR spans [0.44, 1.00] and on Infection [0.15, 0.85]. An AP of 0.995 on three boxes is arithmetic, not evidence. Worse, BDC/BDR has *zero* instances in the test split, so the held-out set cannot score one of the six classes at all.

The protocol the data supports. Pooling all 231 radiographs and rotating through k folds evaluates on every image instead of 10% of them. Splits must be grouped by source radiograph so augmented copies never straddle a fold boundary, and assignment must be rarity-aware or the scarce classes vanish from some folds. Our implementation assigns groups greedily, rarest class first, to the fold most starved of that group’s classes. On $k = 5$ this yields 46–47 source radiographs per fold with *every* class present in *every* fold, and raises pooled evaluation support for BDC/BDR from 3 instances to 72—a $24\times$ increase—at a cost of five trainings instead of one.

4 Method

Models. Three detector families were fine-tuned from COCO-pretrained weights under an identical recipe: YOLOv8n [3], YOLOv10s [11] and YOLOv12m, each for 250 epochs at 640×640 , batch 8, with mosaic, MixUp, copy-paste, HSV jitter, and geometric augmentation (rotation 10° , shear 5° , scale 0.5, translate 0.1, horizontal flip 0.5, vertical flip 0.2).

Metrics. We compute average precision ourselves rather than reading it off the training framework, because bootstrap intervals require recomputing the metric on hundreds of re-samples and re-running inference each time is prohibitive. Inference runs once and is cached per image; every reported number, including every bootstrap replicate, is derived from that cache. The implementation follows the COCO convention (greedy confidence-ordered matching, all-point interpolated area) and is pinned by unit tests against Ultralytics’ own `ap_per_class` to within 0.02 AP across randomised inputs.

Precision, recall and F1 are reported *at the deployed confidence threshold* (0.25), not as curve summaries, because the operating point is what a clinician sees. Confidence intervals come from a percentile bootstrap resampled over *images* rather than boxes, since boxes within one radiograph are correlated and a box-level bootstrap would understate the interval.

Error decomposition. Every false positive above threshold is assigned to exactly one of: *classification* ($\text{IoU} \geq 0.5$ with ground truth of another class), *localization* ($0.1 \leq \text{IoU} < 0.5$, same class), *both, duplicate* (a correct object already claimed by a higher-scoring detection), or *background* ($\text{IoU} < 0.1$ with everything). Ground truth with no matching detection is counted separately as *missed*.

Explanation. We provide Eigen-CAM (gradient-free, and therefore stable across the three head designs) and a class-conditional Grad-CAM whose scalar target is the summed logits of one class over the top- k anchors. Maps are scored by *pointing-game energy*—the fraction of saliency mass inside the predicted boxes, reported against the box area fraction that a uniform map would achieve—and by *deletion AUC*, the area under the target-class score as the highest-saliency pixels are progressively blanked, reported against a random-order baseline.

5 Results

Per-class breakdown. Table 3 decomposes the headline figure. Two things become visible that the aggregate hid. First, the three classes driving the high mean—Impacted, Fractured, BDC/BDR—are precisely the three with the least support; the two classes with the most evidence behind them, Caries and Healthy, sit at 0.85 and 0.84. Second, the ranking between YOLOv10s and YOLOv12m depends entirely on whether the low-support classes are included. On all six classes YOLOv10s wins by 0.011; on the three measurable classes YOLOv12m wins by 0.013. Neither gap is meaningful without repeated runs—which is the point.

Cost. Table 4 pairs accuracy with wall-clock training time. The largest model in the comparison is the slowest and not

Class	N	v8n	v10s	v12m
Caries	14	0.781	0.849	0.875
Healthy	67	0.734	0.837	0.850
Impacted	18	0.978	0.995	0.995
Infection*	4	0.750	0.870	0.788
Fractured*	9	0.973	0.995	0.968
BDC/BDR*	3	0.671	0.995	0.995
mAP@0.5 (all)		0.815	0.923	0.912
mAP@0.5 (N$\geq$10)		0.831	0.894	0.907

Table 3: Per-class AP@0.5 on the validation split. Classes marked * have fewer than ten instances and are not reliable estimates. Restricting the mean to the three measurable classes reverses the ranking of YOLOv10s and YOLOv12m, which is the practical cost of quoting one blended number.

Model	Best ep.	mAP@0.5	mAP@.5:0.95	Train
YOLOv8n	198	0.795	0.480	30 min
YOLOv10s	214	0.928	0.611	72 min
YOLOv12m	240	0.909	0.625	332 min

Table 4: Cost against accuracy, single run each, same GPU. YOLOv12m costs $4.6\times$ YOLOv10s and does not beat it at IoU 0.5, although its higher mAP@0.5:0.95 indicates tighter boxes on the objects it does find. “Best” is a maximum over epochs on the validation split and is therefore optimistically biased; it is not a held-out estimate.

the most accurate at IoU 0.5. Its advantage at mAP@0.5:0.95 (0.625 vs. 0.611) suggests the extra capacity buys box regression precision rather than detection recall—a distinction that matters for a triage tool, where finding the lesion is the task and a slightly loose box is not a clinical failure. Figure 1 shows all three trajectories; none has plateaued by epoch 250, and the run-to-run oscillation of ± 0.03 mAP between adjacent epochs is itself larger than the gap between the two larger models.

Error analysis. The confusion structure of the YOLOv10s model is unusually clean: inter-class confusion accounts for approximately 1% of predictions. Essentially every error is a missed detection or a background false positive. Recall by class is Impacted 1.00, BDC/BDR 1.00, Fractured 0.89, Caries 0.79, Healthy 0.64, Infection 0.50; the complement of each is, in every case, ground truth assigned to background rather than to a wrong class. Of the false positives raised on background, 82% are labelled Healthy, 9% Caries and 9% BDC/BDR.

The interpretation is specific and actionable. The classification head is not the bottleneck—given a proposal, the model names it correctly. The bottleneck is proposal recall, concentrated on Infection (half of all periapical radiolucencies are never proposed) and on Healthy, where the overwhelming instance count paradoxically coincides with the worst recall-to-background ratio, consistent with the detector treating unremarkable tooth regions as background. This points at loss weighting, confidence thresholding and input resolution rather than at the label taxonomy or the classifier.

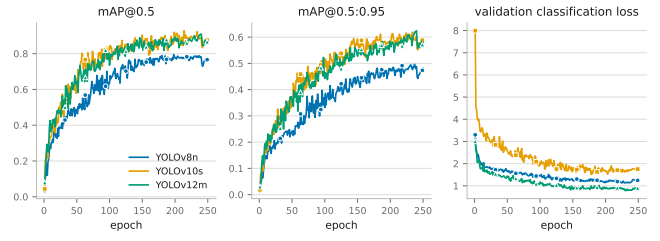


Figure 1: Training trajectories. The epoch-to-epoch oscillation on a 23-image validation split is comparable in size to the differences being compared.

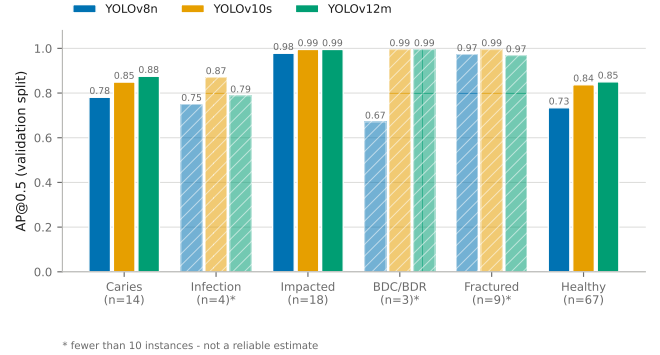


Figure 2: Per-class AP@0.5. Hatched bars mark classes with fewer than ten validation instances. The tallest bars in the figure rest on the least data.

6 Protocol and reproduction

The audit above is a description of a problem; the repository accompanying this report implements the fix in four command-line tools: `dataset.py` (prepare, audit with leakage check, grouped rarity-aware fold construction), `model.py` (multi-seed training, ablation, aggregation), `evaluate.py` (per-class metrics with bootstrap intervals, head-to-head comparison) and `evaluate_new.py` (error decomposition, threshold sweep, scored saliency). The ablation grid is declared in YAML as deltas from the baseline recipe—eleven augmentation and schedule variants plus four architectures—and every variant is reported as mean $\pm$ standard deviation over seeds with Welch’s t -test and Cohen’s d against the baseline, so that “moved the needle” and “within noise” are verdicts rather than adjectives. At ~ 72 min per YOLOv10s run the full grid is roughly 60 GPU-hours; a 120-epoch first pass over three seeds costs about a third of that and preserves the ranking of variants.

The hypotheses attached to each ablation are stated in advance in the config. Two are worth flagging: mosaic augmentation composites four radiographs into one, destroying the fixed global anatomy of a panoramic view, and may therefore hurt on this modality despite being a reliable win on natural images; and HSV jitter on a greyscale modality can only perturb value, so its hue and saturation components are expected to be exactly inert.

7 Limitations

The single largest limitation is data. 231 radiographs from one source, with one annotator and no inter-rater agreement statistics, cannot support a claim about clinical performance in any population. The **Healthy** class is not a condition but a background label, which makes the six-class mean a mixture of two different tasks. No calibration analysis is reported: the confidence values the interface displays have not been checked against empirical accuracy, and should not be read as probabilities. The system is a research artefact and a demonstration; it is not a diagnostic device and must not be used as one.

8 Conclusion

Decomposing a single mAP@0.5 of 0.92 into its six components showed that the classes contributing most to it are the ones with the least evidence behind them, that the ranking of the two leading architectures inverts depending on which classes are counted, and that the model’s residual error is almost entirely missed detections rather than misclassification. The corrective is not a better backbone but a protocol that the dataset can actually support: grouped cross-validation, repeated seeds, per-class intervals, and explanations scored rather than displayed.

References

- [1] Mahmud Ahmed and Md. Ashiqur Rahman. Dental OPG x-ray dataset. *Mendeley Data*, 2023.
- [2] Daniel Bolya, Sean Foley, James Hays, and Judy Hoffman. TIDE: A general toolbox for identifying object detection errors. In *European Conference on Computer Vision (ECCV)*, 2020.
- [3] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics YOLOv8. <https://github.com/ultralytics/ultralytics>, 2023.
- [4] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: Common objects in context. *European Conference on Computer Vision (ECCV)*, 2014.
- [5] Mohammed Bany Muhammad and Mohammed Yeasin. Eigen-CAM: Class activation map using principal components. In *International Joint Conference on Neural Networks (IJCNN)*, 2020.
- [6] Karen Panetta, Rahul Rajendran, Aditya Ramesh, Shreyas Rao, and Sos Agaian. Tufts dental database: A multimodal panoramic x-ray dataset for benchmarking diagnostic systems. *IEEE Journal of Biomedical and Health Informatics*, 26(4):1650–1659, 2021.
- [7] Vitali Petsiuk, Abir Das, and Kate Saenko. RISE: Randomized input sampling for explanation of black-box models. In *British Machine Vision Conference (BMVC)*, 2018.
- [8] David Picard. Torch.manual_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision. In *arXiv:2109.08203*, 2021.
- [9] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *International Conference on Computer Vision (ICCV)*, 2017.
- [10] Gaël Varoquaux and Veronika Cheplygina. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digital Medicine*, 5(48), 2022.
- [11] Ao Wang, Hui Chen, Lihao Liu, Kai Chen, Zijia Lin, Jungong Han, and Guiguang Ding. YOLOv10: Real-time end-to-end object detection. *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.